MSC: A Benchmark Dataset for Machine Learning as a Service (MLaaS) Composition

Abstract. Machine Learning as a Service (MLaaS) service composition has emerged as an active research domain that combines multiple machine learning (ML) services to meet complex service requirements. Despite growing interest, current research lacks high-quality datasets that capture ML characteristics like model specifications and diverse QoS attributes. Furthermore, the lack of standardized service requirements and composition solutions limits evaluation of composition approaches. Existing service composition datasets are outdated and unsuitable for MLaaS composition research. To address these limitations, we introduce the MLaaS service composition (MSC) dataset, the largest known ML service composition dataset to date, consisting of 25,900 MLaaS services from various providers. The dataset contains 20,000 unique service requirements that cover diverse composition scenarios with different objective functions, constraints, preference weights, and the corresponding high-quality composition solutions. To support further research in MLaaS service composition, we develop a benchmark based on the proposed dataset to systematically evaluate different composition strategies and algorithms. In addition, the dataset enables broader applications such as service recommendation, classification, and QoS prediction. The dataset and implementation are publicly available at: https://anonymous.4open.science/r/MLaaS_Dataset_Gen-F3C4/

Keywords: MLaaS Datasets · Service Composition · Service Selection

1 Introduction

Machine Learning as a Service (MLaaS) is a cloud-based service paradigm that provides infrastructure to build, train, and deploy ML models [1]. Major providers such as *Amazon*, *Google*, *Microsoft*, and *Hugging Face* offer a wide range of MLaaS solutions, covering tasks such as classification and predictive modeling. For example, *Hugging Face*[1] hosts more than *two million* ML services and datasets through its platform. These services are widely adopted in data-driven applications across domains such as healthcare, industrial, and digital platforms [2]. Relying on a single MLaaS service is often insufficient to meet complex service requirements. For example, in healthcare, a single MLaaS may not support both patient activity recognition and health risk prediction. In such cases, integrating multiple services is required to meet both functional and QoS requirements.

MLaaS service composition is an emerging paradigm that integrates multiple ML services into a composite service to meet complex application requirements. It is broadly categorized into sequential and collaborative composition, as illustrated in Fig. 1. In a sequential composition, services are organized in a workflow in which each service performs a specific task and passes its output to the next service [3]. For example, a healthcare application may convert a medical image into text and use it to generate a clinical summary for further analysis. In contrast, a collaborative composition combines multiple MLaaS services into a unified service, where each service contributes different functional capabilities or QoS characteristic [4, 5]. For example, in a human activity recognition (HAR) application, one service can detect walking and running, while another service may recognize sitting activities. Composing these services results in a unified HAR service with broader functionality and improved overall performance.

[1] https://huggingface.co/docs/hub/main/en/index

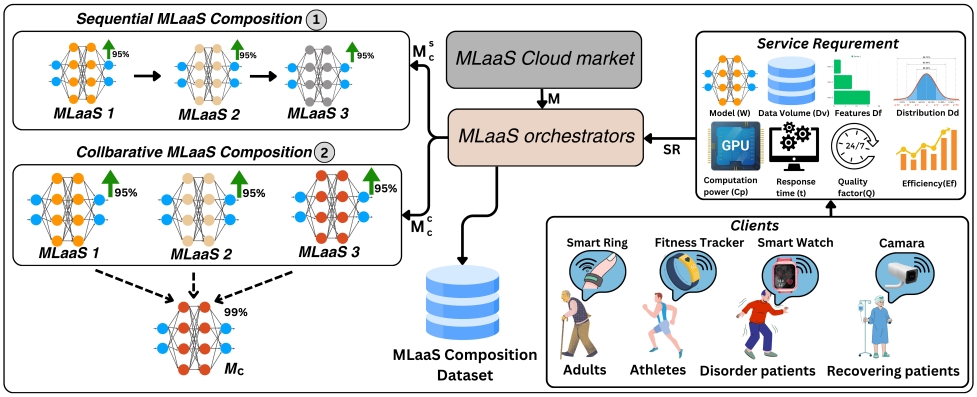


Fig. 1: MLaaS Composition Dataset Motivation Scenario

Let us assume an MLaaS composition environment (see Fig. 1), where clients submit service requirements to MLaaS orchestrators to identify composition solutions. Orchestrators extract service information from cloud marketplaces to perform tasks such as service selection, recommendation, and composition. However, most MLaaS service information remains incomplete and unclear because providers do not expose all model characteristics and diverse QoS attributes required for composition. In addition, evaluating MLaaS composition approaches requires benchmark datasets containing service requirements and composition solutions. To the best of our knowledge, there is no publicly available MLaaS dataset that provides these capabilities. To address this limitation, we introduce the *MLaaS Service Composition Dataset*.

Existing studies in service computing have developed benchmark datasets such as QWS [6], WSDREAM [7], and OWLS-TC2 [8]. These datasets are designed for traditional web and cloud services, are now outdated and are not applicable for MLaaS composition. Unlike traditional services, MLaaS services involve distinctive attributes related to training data specifications (e.g., modality and data distribution), model specifications (e.g., model type and parameters) and QoS aspects (e.g., evaluation metrics, explainability, execution cost, and resource usage). These attributes are critical for tasks such as service selection, recommendation, and composition, yet are largely absent in existing datasets. A recent study introduced the HSC dataset for large-scale AI service composition [3]. However, it primarily focuses on sequential composition scenarios and lacks important MLaaS-specific attributes, including model specifications, training data specifications, and QoS aspects. Furthermore, it does not support collaborative MLaaS composition scenarios due to limited attributes. Therefore, existing datasets fail to effectively support both sequential and collaborative MLaaS composition scenarios under realistic environments.

To address these challenges, we introduce a comprehensive dataset specifically designed for MLaaS composition research. Unlike existing datasets restricted to a single platform or a narrow set of models, the proposed dataset includes heterogeneous MLaaS services integrating both pre-trained and fine-tuned models collected from *5 providers*. The resulting MSC dataset comprises *25,900* MLaaS services spanning *12 ML task* families, including classification, regression, detection, clustering, segmentation, and natural language processing (NLP). To reflect real-world usage, the framework incorporates a service requirement generator that produces application-driven requirements by modeling functional needs, QoS constraints, and user preferences for both sequential and collaborative composition scenarios. For each generated request, the framework derives corresponding composition solutions using a *multi-armed bandit (MAB)-based MLaaS composability algorithm* to identify feasible and high-quality service combinations under varying conditions. The resulting service requirements and associated composition solutions are evaluated using multiple composability models and optimization strategies to estimate baseline performance. This design provides

a structured mapping between service requirements and composition outcomes, enabling researchers to assess and compare different composition and optimization techniques under consistent constraints. As a result, the dataset serves as a benchmark resource for evaluating MLaaS composition solutions in diverse scenarios. Our key contributions are summarized as follows.

- We propose the *MLaaS Service Composition Dataset (MSC)* workflow designed to support both sequential and collaborative MLaaS composition scenarios.
- We construct a large-scale MSC dataset containing more than *25,900* MLaaS services from multiple providers evaluated on benchmark datasets.
- We generate *20,000* service requirements with diverse objectives, QoS constraints, preference weights, and corresponding composition solutions.
- We develop a MAB-based MLaaS composability algorithm to generate benchmark composition solutions for benchmarking MLaaS composition approaches

2 Related Work

MLaaS composition has recently emerged as a new paradigm within the broader field of service composition [5]. Recent studies on MLaaS composition [4] and MLaaS federation [2] have demonstrated their potential to support advanced IoT applications. However, due to the lack of publicly available datasets and privacy constraints in MLaaS environments, most studies rely on individually developed ML models and manually collected service characteristics for composition evaluation. This leads to incomplete service information, limited composition diversity, and expensive evaluation processes for testing different composition scenarios [9]. Traditionally, service composition research has relied on datasets containing a limited number of fixed services. Early foundational resources, such as the QWS dataset , provided QoS measurements for 2,507 web services annotated across nine QoS parameters [6]. Similarly, the WS-DREAM dataset later expanded the scale to 21,358 web services, enabling QoS analysis across dimensions such as response time, throughput, and failure probability [7]. Another service composition work introduced two benchmark datasets: OWLS-TC, containing 1,083 services and 42 queries, and SAWSDL-TC, including 1,080 services and 42 requests, both annotated with multiple ontologies for semantic service retrieval evaluation [8]. In addition, the Web Services Challenge (WSC) benchmark series supported the evaluation of functional and QoS-aware service composition using WSDL, OWL, and WSLA specifications [10].

Recently, the HSC dataset was introduced to support AI service composition research by providing 17,536 AI services across 20 ML tasks for sequential composition [3]. Although the dataset significantly improves the scale of AI service composition research, it still has several limitations, including limited QoS attributes such as explainability and cost, insufficient internal ML parameters, and no support for collaborative MLaaS composition scenarios. Table 1 highlights the gaps in MLaaS-specific attributes and composability support, limiting the applicability of existing datasets to realistic MLaaS composition environments.

Table 1: Existing datasets for MLaaS service composition research.

Datasets	MLaaS specific	Functional attributes	QoS attributes	Internal parameters	XAI scores	Composability support
QWS [6]	❍	❍	●	❍	❍	❍
WS-DREAM [7]	❍	❍	●	❍	❍	❍
OWLS-TC [8]	❍	●	❍	❍	❍	❍
IMLaaS [5]	●	●	◗	❍	❍	◗
HSC [3]	◗	●	●	◗	❍	◗
MSC	●	●	●	●	●	●

Note: ● supported, ◗ partially supported, ❍ not supported.

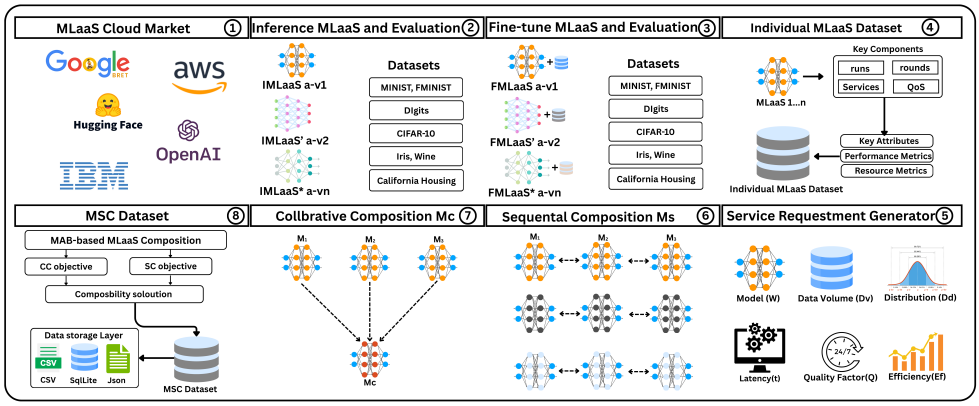


Fig. 2: Overall workflow of the MLaaS service composition dataset collection

3 MLaaS Service Composition Dataset

In this section, we discuss the MSC dataset collection methodology, including MLaaS services, service requirements, composition scenarios, and MLaaS composition. Figure 2 illustrates the overall workflow of the MSC dataset collection. The workflow first collects heterogeneous inference-based and fine-tuned MLaaS services from multiple providers and extracts their attributes using benchmark datasets. It then generates realistic service requirements for sequential and collaborative composition scenarios. Finally, a MAB-based MLaaS composability algorithm is applied to generate composition solutions and construct the final MSC dataset for benchmarking and evaluation.

3.1 MLaaS Service

The rapidly growing MLaaS ecosystem offers diverse repositories and providers, enabling new applications through the composition of MLaaS services. To support realistic composition scenarios, we focus on services that are practically usable and composition-ready. We selected *12 ML task* families spanning major domains such as natural language processing, computer vision, predictive modeling, clustering, regression, detection, and segmentation, as summarized in Table 2. To reflect realistic MLaaS deployment scenarios, we construct services using both inference-ready and fine-tuned models. Since commercial MLaaS providers typically expose only limited model-level information to clients, fine-tuning enables the derivation of representative internal model characteristics and training behaviors of original services. To evaluate these services, we used widely adopted benchmark datasets associated with each task category to capture observable performance across functional and QoS attributes. On *May 01, 2026*, we collected approximately *1,500* top-ranked models across the selected tasks, resulting in more than *25,900* services, as shown in Table 2. Each service was described using training data specifications, internal model parameters, and QoS-related attributes such as accuracy, latency, reliability, cost efficiency, and explainability score, along with more than *20* additional MLaaS-specific attributes. Table 3 summarizes the descriptions of key service attributes used in the dataset.

Unlike traditional dataset construction approaches that rely only on static metadata, we also evaluated runtime service behavior by repeatedly measuring response times under different interaction conditions to capture service-side variability. Moreover, we introduced an explainability score attribute by analyzing how MLaaS service responses change under diverse input variations to measure their sensitivity to input perturbations. Services exhibiting stable and less sensitive behavior under changing inputs were considered highly explainable , whereas highly sensitive services are less explainable. Finally, to ensure that functional and QoS attributes could be compared within a unified scale for downstream composition analysis, we applied min-max normalization to all applicable parameters, except those already bounded in the range of

Table 2: MLaaS benchmark tasks, sources, datasets, and number of services

Task Family	**Source**	**Evaluated Dataset**	#
Fill Mask	HF, BERT	WikiText-2	3,288
Text Classification	HF, BERT	SST-2, AG News, IMDb	3,008
Tabular Regression	sklearn, XGBoost	California Housing	2,998
Sentence Similarity	HF, MiniLM	STS-B, MRPC	2,377
Image Classification	Keras, sklearn, ViT	CIFAR-10, MNIST, FMNIST	3,978
Object Detection	HF, DETR	COCO	2,521
Token Classification	HF, BERT	CoNLL-2003, WNUT-17	2,045
Image Segmentation	HF, SegFormer	ADE20K	1,174
Clustering	sklearn, KMeans	Iris, Wine	1,000
Regression	sklearn	Diabetes, California House	1,000
Text Generation	HF, GPT-2, T5	WikiText-2	2,011
Text-to-Text Gen	HF, T5	XSum	500
Total			**25,900**

Table 3: Formal MLaaS service attributes, symbols, and descriptions

Attribute	**Symbol**	**Description**
Data distribution	Δ	Data distribution type, such as IID or non-IID.
Dataset size	D_n	Number of training samples.
Model parameters	w	Model weights and bias.
Accuracy	A	Final task performance score.
Latency	L	Average inference response time.
Reliability score	R	Consistency of service performance.
Compute time	T	Average computation time.
Resource cost score	C	Operational resource cost.
Cost efficiency	CE	Performance per unit cost.
Model size	M	Number of model parameters.
Explainability score	X	Model interpretability measure.

0 to 1, such as reliability and explainability scores. As a result, the normalized dataset provides a consistent representation of service characteristics while preserving the relative structure of the original measurements.

3.2 Service Requirement

Service requests are a key aspect of service composition, where clients define their functional, QoS requirements, preference weights, and constraints. Due to the difficulty in collecting real-world MLaaS service requests an a scale, we generate these requests based on established machine learning service selection principles reported in prior studies [11]. These principles reflect application-driven service selection preferences commonly considered in MLaaS and AI-based systems, including the prioritization of attributes such as accuracy, latency, reliability, explainability, and resource efficiency. For example, healthcare applications typically prioritize *high accuracy* and *low latency*, whereas financial applications emphasize *high reliability* and *explainability*.

Existing studies represent MLaaS service requirements using four key dimensions: functional requirements, QoS requirements, preference weights, and constraints. Functional requirements describe the expected service capabilities, including model type (e.g., CNN-based), supported data modalities (e.g., sensor, image, multimodal), and training data characteristics such as data volume, features, labels, and distribution. QoS requirements represent performance-related properties, including accuracy, latency, reliability, explainability, and resource efficiency. Preference weights are used to prioritize specific attributes based on application requirements, such as emphasizing higher accuracy or lower latency. Constraints define acceptable operational limits for

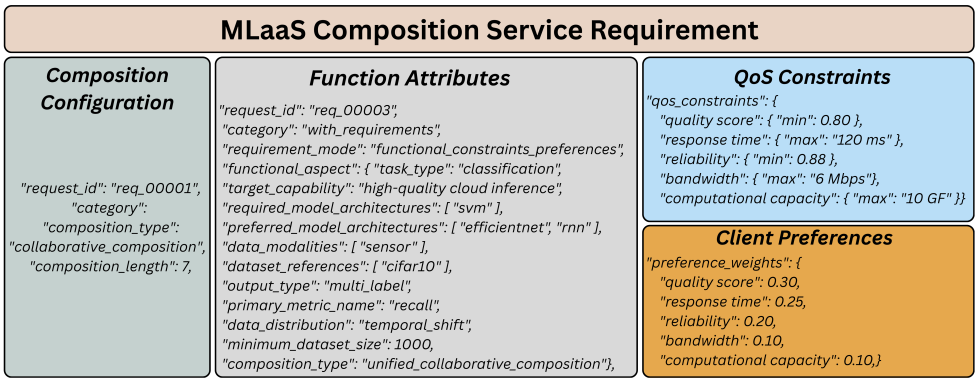


Fig. 3: An example of a MLaaS Composition Service Requirement

service selection, such as minimum accuracy thresholds or maximum latency bounds. Using these dimensions, we generate realistic MLaaS service requests for both collaborative and sequential composition scenarios. In collaborative composition, service requests mainly focus on compatibility across these four dimensions, whereas sequential composition additionally considers workflow dependencies and the ordering of services within the composition pipeline. Figure 3 illustrates a fundamental example of an MLaaS service request structure.

We design an MLaaS service requirement generation algorithm (SR) for both sequential and collaborative composition (*Algorithm 1*). The algorithm constructs requests by selecting preference profiles and ML selection principles from the MLaaS dataset, while dynamically determining the composition type as either sequential or collaborative. Each generated request includes functional requirements (F), QoS requirements (QoS), preference weights (PW), and constraints (C), which are updated based on predefined selection principles to reflect realistic MLaaS selection behavior. For sequential composition, the algorithm derives workflow structures and service ordering, whereas collaborative composition determines compatible service group sizes. Finally, objective functions and constraints are defined to construct realistic service requirements aligned with practical performance and quality requirements.

Algorithm 1 MLaaS Service Requirement Generation

Require: Dataset MD, profiles R, principles P, requests N
Ensure: Service requests SR

$SR \leftarrow \emptyset$
for $i = 1$ to N **do**
  Select profile $r_i \in R$ and principle $p_i \in P$
  Select composition type $CT_i \in \{sequential, collaborative\}$
  Extract F_i, QoS_i, PW_i, and C_i from r_i
  Randomly select attributes from $F_i \cup QoS_i \cup PW_i \cup C_i$
  for all selected attribute a **do**
    Apply principle p_i and update a within valid bounds
  end for
  if $CT_i = sequential$ **then**
    Derive workflow W_i from MD
  else
    Derive collaborative size K_i from MD
  end if
  Define objective function OF_i
  $sr_i \leftarrow \langle F_i, QoS_i, PW_i, C_i, CT_i, W_i/K_i, OF_i \rangle$
  $SR \leftarrow SR \cup \{sr_i\}$
end for
return SR

Table 4: MLaaS composability function for sequential and collaborative composition

Attributes	**Sequential composition (S_c)**	**Collaborative composition (C_c)**
Data parameters (Δ, D_n)	$\frac{1}{n}\sum_{i=1}^{n} D_i$	$1 - \frac{1}{m}\sum_{j=1}^{m} \frac{\max_{i=1}^{k}(\Delta_{ij}) - \min_{i=1}^{k}(\Delta_{ij})}{\sum_{i=1}^{k}\Delta_{ij}}$
Model parameters (w, Loss)	$1 - \frac{1}{n}\sum_{i=1}^{n} Loss_i$	$\frac{1}{k}\sum_{i=1}^{k} \frac{\langle \nabla w_i, \nabla w_c \rangle}{\|\nabla w_i\| \|\nabla w_c\|}$
Computation time (T)	$\frac{1}{n}\sum_{i=1}^{n}(1 - T_i)$	$1 - \frac{1}{k}\sum_{i=1}^{k} \|T_i - \mu_T\|$
Latency (L)	$\frac{1}{n}\sum_{i=1}^{n}(1 - L_i)$	$\min\left(1, \frac{L_{\max}}{\sum_{i=1}^{k} L_i}\right)^{\alpha}$
Accuracy (A)	$\frac{1}{n}\sum_{i=1}^{n} A_i$	$1 - \frac{1}{k}\sum_{i=1}^{k} \|A_i - \mu_A\|$
Reliability score (R)	$\sqrt[n]{\prod_{i=1}^{n} R_i}$	$1 - \frac{1}{k}\sum_{i=1}^{k} \|R_i - \mu_R\|$
Explainability score (X)	$\frac{1}{n}\sum_{i=1}^{n}(1 - X_i)$	$1 - \frac{1}{k}\sum_{i=1}^{k} X_i$
Cost (C)	$\frac{1}{n}\sum_{i=1}^{n}(1 - C_i)$	$1 - \frac{1}{k}\sum_{i=1}^{k} C_i$
Objective function	$SC = \sum_{j=1}^{m} \lambda_w^j S_c^j$	$CC = \sum_{j=1}^{m} \lambda_w^j C_c^j$

3.3 MLaaS Composition solution

To evaluate a MLaaS composite solution, we define the overall composability objective functions for sequential and collaborative composition over a candidate service combination $S_k = \{M_1, M_2, \ldots, M_k\}$ as follows:

$$\max\ SC(S_k) = \sum_{j\in\{1,2,..8\}} \lambda_w^j S_c^j(S_k), \quad \max\ CC(S_k) = \sum_{j\in\{1,2,..8\}} \lambda_w^j C_c^j(S_k) \tag{1}$$

where $S_c^j(S_k)$ and $C_c^j(S_k)$ denote the jth composability scores for sequential and collaborative composition over the service combination S_k, respectively, and λ_w^j represents the corresponding preference weight defined by the service requirement.

Sequential composition focuses on aggregating the individual functional and QoS capabilities of MLaaS services across workflow stages, whereas collaborative composition evaluates the compatibility and interaction behavior among multiple services. The detailed composability formulations for sequential and collaborative composition are presented in Table 4. The formulation is designed to capture the composability characteristics identified in existing studies on machine learning and collaborative learning environments, where both functional and QoS attributes influence the effectiveness of service composition [5, 11]. Functional characteristics such as data heterogeneity, data volume, model parameter alignment, and training loss behavior are considered to evaluate learning compatibility among MLaaS services. In addition, QoS-related attributes including computation time, latency, accuracy, reliability, explainability, and cost are incorporated to assess the operational quality and deployment feasibility of the composed solution. The formulation also supports application-driven service selection through preference weights, allowing different functional and QoS criteria to be prioritized according to service requirements and deployment objectives.

To ensure that the selected service combination satisfies the requested constraints, a penalty term is incorporated to capture deviations from the target functional and QoS requirements. Let $O(S_k)$ denote either $SC(S_k)$ or $CC(S_k)$. The penalty-aware composability objective is then defined as:

$$\max\ \hat{O}(S_k) = O(S_k) - \sum_{j\in\mathcal{R}} \eta_j \left| q_j(S_k) - q_j^r \right|^2 \tag{2}$$

Algorithm 2 Multi-Armed Bandit based MLaaS Composition

Input: $\langle SR, MD, SA, K, train \rangle$
Output: $bestS_k, bestCS$ ▷ Best service combination and best composability score
Generate candidate service combinations Ω from MD and SA based on SR with size K
Initialize $r_{\text{est}}(S_k) \leftarrow 0,\ n(S_k) \leftarrow 0$ for all $S_k \in \Omega$
$bestCS \leftarrow -\infty,\ bestS_k \leftarrow$ None
for $t = 1$ to $train$ **do**
  for all $S_k \in \Omega$ **do**
    $ucb(S_k) \leftarrow \begin{cases} \infty, & n(S_k) = 0 \\ r_{\text{est}}(S_k) + \sqrt{\frac{2 \log t}{n(S_k)}}, & otherwise \end{cases}$
  end for
  $S_k^* \leftarrow \arg\max_{S_k \in \Omega} ucb(S_k)$
  if SR is sequential composition **then**
    $cs \leftarrow SC(S_k^*)$ ▷ Sequential composition
  else
    $cs \leftarrow CC(S_k^*)$ ▷ Collaborative composition
  end if
  $n(S_k^*) \leftarrow n(S_k^*) + 1$
  $r_{\text{est}}(S_k^*) \leftarrow r_{\text{est}}(S_k^*) + \frac{cs - r_{\text{est}}(S_k^*)}{n(S_k^*)}$
  if $cs > bestCS$ **then**
    $bestCS \leftarrow cs,\ bestS_k \leftarrow S_k^*$
  end if
end for
return $bestS_k, bestCS$

where $\mathcal{R}$ denotes the set of constrained service request attributes, $q_j(S_k)$ represents the aggregated value of the jth attribute achieved by the composed MLaaS solution, q_j^r denotes the corresponding target value specified in the service request, and η_j represents the penalty coefficient associated with the jth constraint. The penalty term reduces the overall composability objective when the selected service combination deviates from the requested functional and QoS requirements.

To evaluate MLaaS composition strategies, the objective is to identify service compositions that achieve optimal composability objective values, where a higher value of $\hat{O}(S_k)$ indicates a better composition. The composition process inherently forms a combinatorial optimization problem, as multiple services with varying capabilities must be selected and combined to satisfy given requirements. For instance, if the composition length is $k = 2$ and the total number of available services is $n = 100$, the number of possible combinations grows significantly, making exhaustive exploration computationally infeasible. The total number of possible combinations can be calculated as $C(n, k) = \frac{n!}{k!(n-k)!}$, which yields 4950 possible combinations, illustrating how rapidly the search space expands. To address this challenge, optimization-based service selection techniques are employed instead of brute-force enumeration. Specifically, a multi-armed bandit (Algorithm 2) is used for relatively smaller search spaces, enabling efficient exploration–exploitation trade-offs. For larger and more complex search spaces, a genetic algorithm with increased population size and iterations is adopted to approximate near-optimal compositions. These optimization strategies enable the discovery of high-quality MLaaS compositions that serve as benchmarks for evaluating different sequential and collaborative composition approaches.

4 Statistical Analysis

The MSC dataset comprises 25,900 publicly available ML services with their corresponding functional and QoS attributes, making it one of the largest publicly accessible datasets for MLaaS composition research. Figure 4 presents a statistical analysis of the generated ML services across task families, model architectures, QoS attributes, and task–architecture relationships.

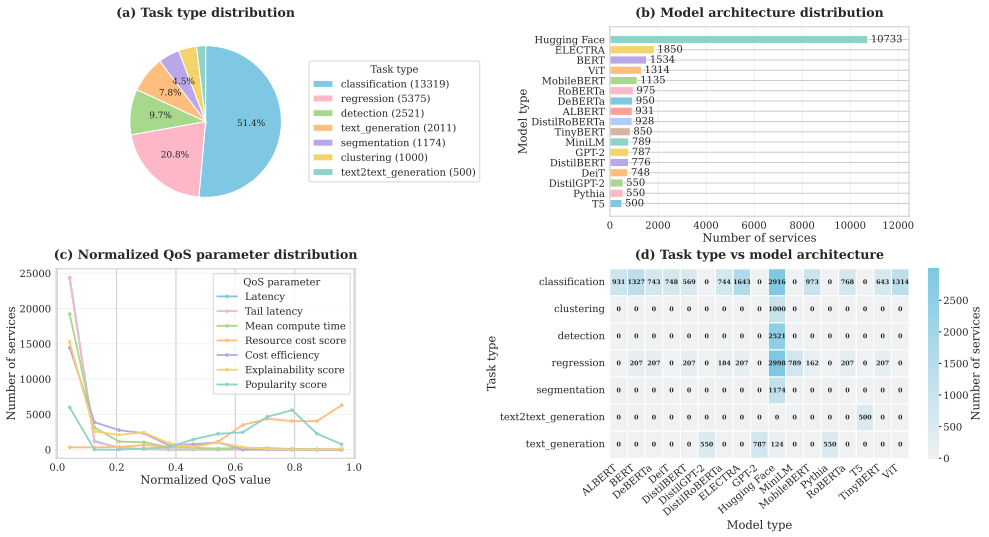


Fig. 4: Statistical analysis of the MLaaS dataset showing a) task-type distribution, b) model architecture distribution, c) normalized QoS parameter distribution, and d) task–architecture associations

Figure 4a illustrates the distribution of major ML task families, including classification, regression, detection, clustering, segmentation, and text-generation tasks. Classification dominates the dataset with more than 13,000 services (51.4%), while the remaining task families provide balanced diversity for heterogeneous composition scenarios. Figure 4b presents the distribution of model architectures. Transformer-based architectures, particularly Hugging Face and BERT-related variants, constitute a major portion of the dataset, alongside other architectures such as RoBERTa, ELECTRA, GPT-based models, and traditional machine learning approaches. This demonstrates the architectural diversity of the collected ML services. Figure 4c shows the normalized QoS parameter distributions. Since QoS attributes are measured at different numerical scales, min–max normalization is applied to project all attributes into a unified range. The resulting distributions reveal variations and trade-offs among latency, computation time, cost efficiency, explainability, and popularity indicators. Finally, Figure 4d illustrates the relationship between task families and model architectures. The heatmap shows that transformer-based architectures are strongly associated with classification and text-oriented tasks, while other task families exhibit comparatively smaller architectural distributions. Overall, these observations confirm that the dataset captures diverse functional, architectural, and QoS characteristics required for evaluating realistic MLaaS composition strategies.

5 Benchmark

To further advance research on MLaaS service composition, we develop a benchmark framework for sequential and collaborative composition using the MSC dataset. The benchmark is designed to support researchers in accurately evaluating and comparing composition solutions under diverse functional and QoS requirements.

5.1 Experiment design

MLaaS Composition Scenario Setup. The MSC dataset comprises 20,000 MLaaS service composition cases, including 10,000 sequential composition scenarios and 10,000 collaborative composition scenarios. The generated composition cases are organized based on different composition sizes, service request configurations, and composability objective functions to support diverse evaluation settings across sequential and collaborative composition environments.

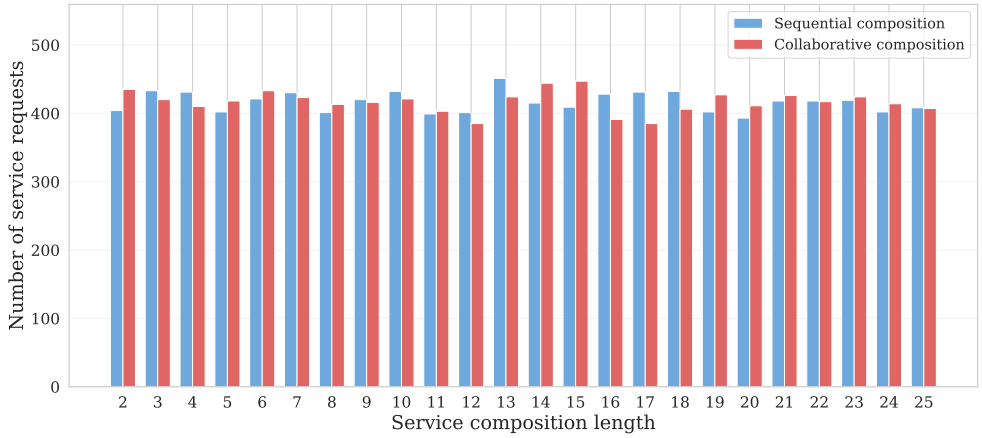


Fig. 5: Service Request Distribution for MLaaS Composition

Baseline techniques In this section, we evaluate the average solution quality of benchmark service composition techniques using the MSC dataset. We measure the average solution quality of MLaaS compositions using the MAB-based MLaaS composability algorithm, which generates best composability solutions for given service requirements. We consider recent service composition techniques widely used in the last 5 years. These include traditional and metaheuristic-based techniques such as Genetic Algorithm [12], Epsilon-Greedy [13], Greedy, and Random Search [14]. In addition, we consider several state-of-the-art optimization approaches, including DAAGA [15], MWOA [16], CSSA [17], SDFGA [18], BPSC-GA [19], and PK-IDPSO [20], which utilize historical service composition information during optimization. We then measure the average solution quality of MLaaS compositions generated using the benchmark composability model to demonstrate that the proposed dataset improves composition quality. All experiments were conducted on an Intel Core i7 machine with 16 GB RAM using Python. The source code is publicly available in our repository[2].

5.2 Evaluation Metrics.

The performance of sequential and collaborative MLaaS composition strategies is evaluated using the average service composition quality score, defined as:

$$ASQ(sol) = \frac{\hat{O}(sol)}{\hat{O}(sol^*)} \tag{3}$$

where sol denotes the generated composition solution and sol^* represents the highest-quality composition solution identified for the corresponding service request. A higher composition quality score indicates a better composition outcome, with values closer to 1 reflecting stronger alignment with the target functional and QoS requirements. To ensure fair comparison among different optimization approaches, all methods are evaluated under comparable experimental settings and composition constraints.

5.3 Experimental Results

We performed three different evaluations to benchmark the proposed MSC dataset. First, we evaluated the proposed MAB-based MLaaS composition algorithm against different optimization approaches under collaborative composition scenarios. Figure 6 shows that traditional approaches such as Random Search and Greedy exhibit lower and less stable performance for longer composition chains, demonstrating their limited applicability in large-scale MLaaS composition environments. In contrast,

[2] https://anonymous.4open.science/r/MLaaS_Dataset_Gen-F3C4/

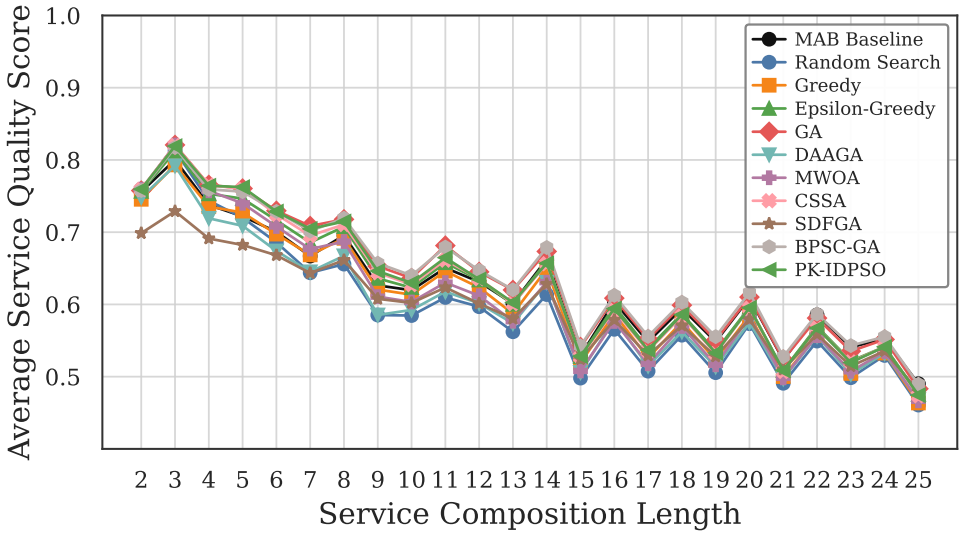


Fig. 6: Average service quality score across different collaborative composition lengths.

metaheuristic-based approaches including GA, DAAGA, MWOA, CSSA, SDFGA, BPSC-GA, and PK-IDPSO generally produce better composition solutions, highlighting the importance of optimization-based exploration in complex composition spaces. The results further demonstrate that the proposed MAB-based composability algorithm provides an effective benchmarking mechanism for evaluating MLaaS composition techniques. Moreover, the relatively small performance gaps across different optimization approaches indicate that the diverse functional and QoS attributes of the MSC dataset support consistently feasible composition solutions across varying service combinations.

Next, we evaluate the average service quality achieved by different optimization strategies under sequential MLaaS composition settings, as illustrated in Fig. 7. Unlike collaborative composition, the results show that sequential compositions maintain consistently higher and more stable quality scores as the composition length increases, although minor fluctuations are observed across different workflow sizes. In addition, the experiment demonstrates that the MSC dataset can effectively capture varying

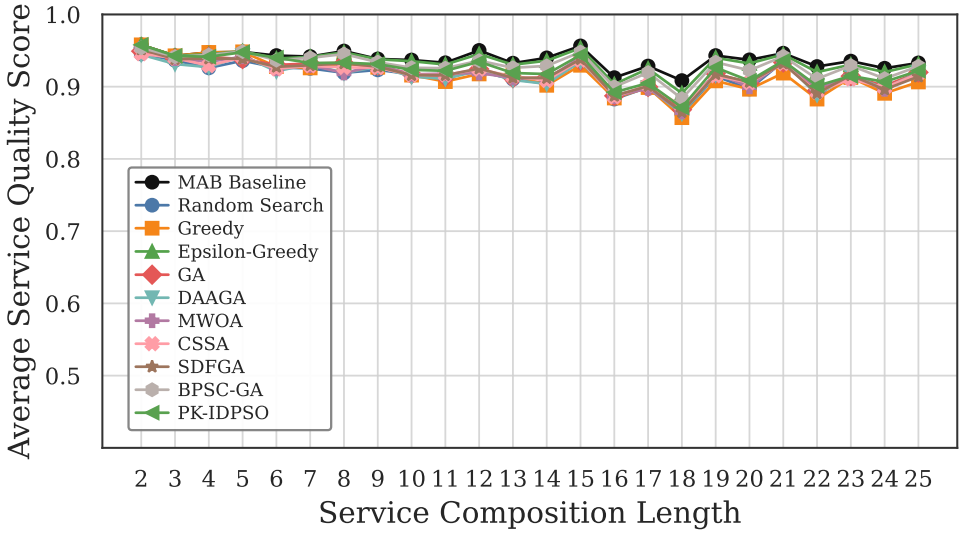


Fig. 7: Average service quality score across different sequential composition lengths.

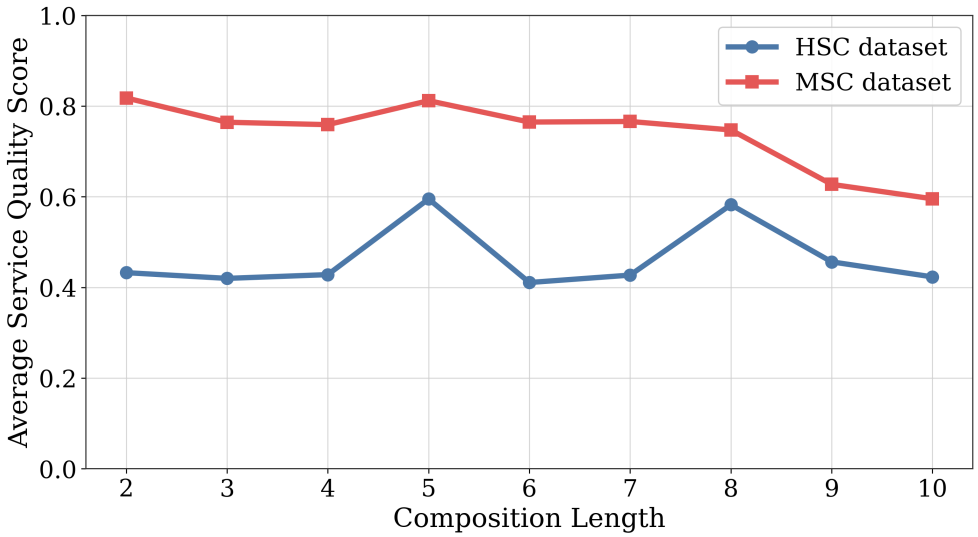


Fig. 8: Benchmark comparison of composition solutions between the MSC and HSC datasets

sequential composition behaviors and provide a realistic benchmark environment for evaluating MLaaS composition strategies across different workflow scales.

Finally, we evaluate the composition solutions of the proposed MSC dataset against the HSC dataset. The HSC dataset has limitations in terms of MLaaS-specific attributes and objective functions, as its composition evaluation is confined to a smaller set of service characteristics. As a result, it may not effectively capture realistic service quality behaviors for diverse MLaaS composition requests. Figure 8 shows that the proposed MSC dataset consistently achieves higher average service quality scores compared to the HSC dataset, demonstrating the effectiveness of the additional functional and QoS attributes in supporting more realistic and composition-aware MLaaS evaluation scenarios.

6 Discussion and Research Implications

The novelty of the proposed MSC dataset enables support for several key tasks in service computing. In this section, we discuss important problems beyond MLaaS composition, including MLaaS service recommendation, service selection and composition, workflow planning and orchestration, service classification, and QoS prediction. We also highlight current limitations and open challenges that remain to be addressed.

6.1 MLaaS Service Recommendation

MLaaS service recommendation focuses on identifying services that best match user requirements by analyzing functional needs and preference patterns. In contrast to composition, recommendation emphasizes functional suitability rather than constructing optimal workflows or modeling inter-service dependencies. User requirements are often expressed in natural language or abstract constraints, requiring mapping to service-level attributes. The proposed *MSC (MLaaS Service Composition Dataset)* supports this task by providing a large set of requirement–service mappings generated through diverse functional and QoS combinations. The dataset includes thousands of requirement instances derived from structured variations of model type, data modality, and QoS preferences. These variations enable evaluation of recommendation approaches under realistic and heterogeneous conditions, where services differ in capability, specialization, and performance trade-offs.

6.2 MLaaS Service Selection and Composition

MLaaS service selection and composition aim to identify and combine multiple services to satisfy complex requirements while considering both functional compatibility and QoS constraints. Unlike recommendation, composition explicitly models inter-service dependencies, workflow feasibility, and overall system performance. The MSC dataset is designed to support this problem by generating *composition-aware service representations*. Each requirement instance is associated with candidate services and feasible compositions derived from functional compatibility (e.g., input–output alignment, data specifications) and QoS attributes (e.g., accuracy, latency, reliability). The dataset incorporates diverse service types, including pre-trained, fine-tuned, and hybrid models, enabling evaluation under realistic conditions. The structured generation of requirement–composition pairs also allows analysis of trade-offs between solution optimality and feasibility.

6.3 MLaaS Workflow Planning and Orchestration

MLaaS orchestration extends beyond composition by incorporating task planning and workflow generation. It involves decomposing high-level requirements into sub-tasks and selecting appropriate services to fulfill each step, forming an executable pipeline. The MSC dataset facilitates orchestration research by providing *requirement-driven workflow structures* generated through Algorithm 1. Each instance includes functional constraints and QoS preferences that guide workflow construction. Unlike traditional datasets that focus only on service selection, MSC explicitly models the planning process by linking requirements to structured workflows and associated objectives. This enables evaluation of orchestration methods that jointly optimize task decomposition, service selection, and execution efficiency.

6.4 MLaaS Service Classification

Service classification categorizes MLaaS services based on their functional descriptions and attributes. This is essential for improving service discovery, selection, and composition efficiency. The MSC dataset includes a diverse collection of services spanning domains such as NLP, computer vision, predictive modeling, and clustering. Each service is described using functional attributes (e.g., model type, data modality, output structure) and QoS characteristics. The scale and diversity of services, including variations from pre-trained and fine-tuned models, provide a strong benchmark for classification tasks. The dataset also reflects realistic challenges such as uneven category distributions and overlapping functionalities.

6.5 QoS Prediction in MLaaS

QoS prediction aims to estimate unknown or future service performance metrics based on observed attributes. This is critical for enabling reliable service selection and composition under uncertainty. The MSC dataset supports QoS prediction by providing multiple QoS attributes for each service, including performance and system-level indicators. These attributes are derived from both platform metadata and empirical evaluation. The normalized QoS parameters ensure consistency across services, enabling the development and validation of predictive models. The dataset also supports analysis of trade-offs between QoS attributes and functional requirements.

7 Limitations and Threats to Validity

The MSC dataset and workflow may be subject to several threats to validity. First, the dataset is constructed using publicly available MLaaS services, model repositories, and benchmark datasets. As a result, some service attributes and metadata may be unavailable due to limitations in publicly advertised specifications. Furthermore, certain

computationally expensive attributes, such as explainability scores, were evaluated using limited sample subsets to maintain practical execution time. Although this may introduce minor deviations from exhaustive evaluation results, repeated experiments showed relatively small variations across runs. In addition, the fine-tuned services in the dataset were generated by further training publicly available pre-trained inference models under controlled evaluation settings. Therefore, the generated services may not fully capture all deployment-specific optimizations used in commercial MLaaS environments. However, the objective of the dataset is not to provide absolute judgments on individual ML services, but to offer a realistic and reproducible benchmarking environment for evaluating MLaaS composition strategies under heterogeneous service and composition settings. To improve reproducibility and reduce experimental bias, the complete dataset generation workflow, evaluation configurations, and service extraction parameters are publicly available through the provided GitHub repository. The framework also allows researchers to customize evaluation settings, extend service attributes, and generate additional MLaaS services based on their own experimental requirements and computational resources.

8 Safety and Ethical Discussion

All services used in the MSC dataset were collected from publicly available MLaaS platforms and service repositories. The evaluation datasets were obtained from widely used public benchmark repositories based on their documented specifications. The MSC dataset only captures observable service characteristics, evaluation behavior, and advertised specifications, without extracting publisher identities, private user information, or sensitive metadata. Furthermore, the reported evaluations are intended solely for benchmarking and comparative analysis within the dataset environment, rather than making definitive judgments about the overall quality or reliability of individual ML services.

9 Conclusion

In this paper, we introduced the MLaaS Service Composition (MSC) dataset, a large-scale benchmark specifically designed to support both sequential and collaborative MLaaS composition research. Unlike existing datasets that are designed for traditional web services and lack MLaaS-specific attributes, the MSC dataset comprises 25,900 MLaaS services collected from five major providers across 12 ML task families, making it the largest known dataset of its kind to date. To enable systematic and fair evaluation, we generated 20,000 unique service composition requirements encompassing diverse objective functions, QoS constraints, preference weights, and corresponding high-quality composition solutions derived using a MAB-based composability algorithm. Experimental results confirm that the dataset effectively differentiates the performance of various composition methods across different composition lengths and scenarios, demonstrating its suitability as a rigorous benchmarking resource. Beyond service composition, the MSC dataset broadly supports additional research directions including service recommendation, service classification, workflow orchestration, and QoS prediction, highlighting its wide practical utility. This dataset serves as a foundational resource that accelerates progress in MLaaS composition research and drives the development of more effective, scalable, and intelligent composition solutions.

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